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Tutorial

D-optimal designs

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Abstract

Many classical symmetrical designs have desirable characteristics, one of which is called D-optimality. The D-optimality concept can also be applied to select a design when the classical symmetrical designs cannot be used, such as when the experimental region is not regular in shape, when the number of experiments chosen by a classical design is too large or when one wants to apply models that deviate from the usual first or second order ones. The D-optimality concept is developed and it is also explained that D-optimality is only one possible criterion to choose a particular design. A few other criteria are also given that complement the information obtained by the D-criterion.

Keywords: Algorithms; Experimental design; D-optimal design; Optimisation

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1. Introduction

Let us consider a case study, namely the optimisation of the separation of six chlorophenols by reversed-phase liquid chromatography (RPLC). Two variables are considered to be important: the pH and the percentage of organic modifier. Table 1 lists the chlorophenols and their pK_A values.

The first step is to establish an experimental region where the chromatograms would have retention times that are neither too short nor too long, that is, where the capacity factor of the chlorophenols would stay within limits such as $1 > k' > 10$ [1,2]. This constraint makes the region irregular in shape and the classical designs, such as factorial, central composite, etc., cannot be used [3]. In our example the region is a polygon with four vertices (Fig. 1). The procedure used to determine the region is explained in [4], and we will start our proposed strategy from the moment where this region is known. Our task is to propose a set of n data points that together will allow us to determine the coefficients in a prediction model.

Some experimental limitations have to be taken into account. Differences smaller than 0.1 pH unit and 1% of organic modifier are considered beyond control and cannot be discriminated. We can therefore consider all possible combinations of the two variables with step sizes of respectively 0.1 pH unit and 1%. The grid drawn in Fig. 1 shows all possible combinations, which from now on we will call *candidate points*. Based on the knowledge of the chromatographer, a model is postulated:

$$\eta = \beta_0 + \beta_1 V_1 + \beta_2 V_2 + \beta_{11} V_1^2 + \beta_{22} V_2^2 + \beta_{12} V_1 V_2 + \beta_{111} V_1^3 \qquad \text{(Model)}$$

where η represents the true value of an analytical response. Because of the experimental error always present in a measurement, it is not possible to have the true value of the analytical response. What one obtains is an estimate represented by Y. In our example the analytical response measured is the logarithm of the capacity factor k'. V_1 and V_2 represent the independent variables (pH and % of organic modifier, respectively) and β_j represents the true model coefficients for variable j. The latter must be estimated from the experiments. The estimates are then denoted by b_j.

The model obtained will then be

$$Y = b_0 + b_1 V_1 + b_2 V_2 + b_{11} V_1^2 + b_{22} V_2^2 + b_{12} V_1 V_2 + b_{111} V_1^3 + e \qquad \text{(Model 1)}$$

where e is an error term.

In experimental design one often applies second-order models to model response surfaces. However the model should be able to describe the behaviour of the retention time of the substances as described by the capacity factor k', in the experimental domain. A polynomial of second order may not fit accurately the retention behaviour in this case, because the evolution of k' as a function of the pH has a sigmoid shape and could probably be better described by a polynomial of a higher degree. Therefore we included a third-order term in the model for variable V_1.

Our objective is to have enough knowledge about the experimental domain to allow us to know the retention times of the substances for every point. To have complete information one should perform all possible experiments in the domain. However, our example contains 257 points and as each point represents one experiment that can be performed, it would be very expensive and it would cost a lot of time to

Table 1
pK_a values of the chlorophenols used in the case study

Chlorophenol	pK_a
2,3	7.71
2,4	7.90
2,5	7.51
2,6	6.78
3,4	8.62
3,5	8.25

perform all of them. It is clear that this is not the best strategy to apply. Another one is to select a set of candidate points that contains enough information about the domain that allow us to predict the retention times of the substances in the points where the experiments were not performed. This set of experiments is called a design.

2. Experimental design matrices

Now that the problem is established, that the candidate points are known and that we know exactly what we are looking for, the questions remain: *How to select a reduced set of points?* and *How many points are needed?* Some mathematical criteria are available to help choosing good sets of candidate points.

Before considering the mathematical criteria, let us explain some of the terminology that will be used in the text.

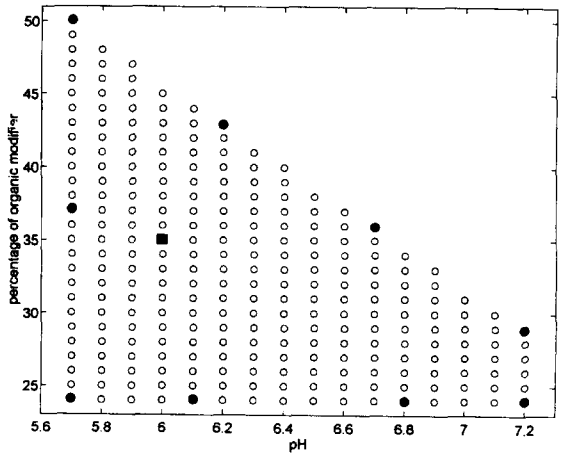


Fig. 1. Experimental region. The grid shows the candidate points. (●) Points chosen for a design with nine experiments; (■) point that permits the validation of the model.

2.1. Matrix of candidate points (ξ_N)

The matrix that contains the candidate points. In this matrix each row represents an experiment and each column a variable and N represents the number of candidate points. In our case the matrix of candidate points is a (257×2) matrix and is represented by (ξ_{257}). From this matrix subsets of candidate points will be chosen. The resulting matrix is also a matrix of candidate points and will be represented by (ξ_n) for a set of n points.

2.2. Model matrix (X)

The model matrix is a $(n \times p)$ matrix where p represents the number of coefficients in the model. The points that are part of it come from ξ_n.

Let us consider an example. Suppose a matrix with six candidate points (ξ_6) described by two variables, the pH and the percentage of organic modifier. Each row of the matrix represents a point and the columns represent the variables.

$$\xi_6 = \begin{pmatrix} 2.1 & 30 \\ 3.5 & 15 \\ 4.9 & 40 \\ 5.1 & 26 \\ 5.7 & 10 \\ 7.0 & 24 \end{pmatrix}$$

Imagine that we need a design with four experiments and that the postulated model to describe the behaviour of a measurement (Y) according to these two variables is

$$Y = b_0 + b_1 V_1 + b_2 V_2^2 \qquad \text{(Model 2)}$$

The corresponding model matrix would have four rows (one for each experiment) and three columns (one for each coefficient of the model). These four experiments are selected from the six in ξ_6. Suppose that we select experiments 1, 2, 3 and 5. The matrix

containing the set of candidate points will be a ξ_4 matrix

$$\xi_4 = \begin{pmatrix} 2.1 & 30 \\ 3.5 & 15 \\ 4.9 & 40 \\ 5.7 & 10 \end{pmatrix}$$

The model matrix constructed from this matrix and model 2 will be

$$\mathbf{X} = \begin{pmatrix} 1 & 2.1 & 900 \\ 1 & 3.5 & 225 \\ 1 & 4.9 & 1600 \\ 1 & 5.7 & 100 \end{pmatrix}$$

The second and third columns in **X** represent respectively the value of the first variable, the pH, and the squared value of the second variable, the percentage of organic modifier, as described in model 2.

The first polynomial term can be seen as

$$b_0 * V_0 = b_0 * 1 \tag{1}$$

where V_0 is the value related to the variables in this term. As it is constant and equal to 1, the first column of **X** consists of ones.

2.3. Optimal design matrix (X^*)

In the example above, the experiments in ξ_4 represent only one of the possible combinations of the six experiments of ξ_6. To select the best combination we use a criterion. In this text, it is in the first instance the D-criterion. The best combination of these points is called *optimal* and the corresponding design matrix is called *optimal design matrix*.

3. Criteria

The more important criteria are: the D-criterion, the G-efficiency criterion and the A-criterion [5–8]. Other criteria such as the E-criterion and Tu-criterion [9–11] are less used so that we will focus on the first three.

3.1. D-criterion

The *D-criterion* was proposed by Smith [12] and states that among all matrices that can be chosen from the matrix of candidate points ξ_N, the one that leads to a model matrix **X** that minimises the determinant of the so-called *dispersion matrix* $(\mathbf{X}'\mathbf{X})^{-1}$, is optimal.

$$\det(\mathbf{X}^{*\prime}\mathbf{X}^{*})^{-1} = \min_{\xi_n \Xi_n} \left(\det(\mathbf{X}'\mathbf{X})^{-1}\right) \tag{2}$$

where $\xi_n \Xi_n$ represents the group of all matrices ξ_n, chosen from ξ_N.

This is equivalent to say that the matrix ξ_n leads to the minimum determinant of the dispersion matrix $(\mathbf{X}^{*\prime}\mathbf{X}^{*})^{-1}$. The elements of the main diagonal in this matrix represent the variance of the model coefficients (b_{ij} where $i = j$) and the other elements (b_{ij} where $i \neq j$) represent their covariance.

Minimising the determinant of the dispersion matrix (Eq. (2)) is equivalent to maximising the determinant of the *information matrix* $(\mathbf{X}'\mathbf{X})$ (Eq. (3)).

$$\det(\mathbf{X}'\mathbf{X}) = \frac{1}{\det(\mathbf{X}'\mathbf{X})^{-1}} \tag{3}$$

One should be aware that in the literature the nomenclature used for the matrix $(\mathbf{X}'\mathbf{X})$ is not consistent. Some authors [13] call this matrix also *dispersion matrix* and others [14,15] call it *information matrix*. In this article we will use this last term.

For constant size of the design, one can state that the higher the determinant of the information matrix, the closer to orthogonality is the dispersion matrix. Orthogonality guarantees the mutual independence of the model coefficients and is especially useful when one is interested in the significance of the model coefficients.

When fitting the model to the experimental data, the experimental error and the error of the model are transmitted to the coefficients. Geometrically, the coefficients and their errors are represented as (hyper)ellipsoids whose axes describe these errors (Fig. 2). In other words, the ellipsoid describes the confidence interval for each coefficient [16]. So, the smaller the axes, the more precise are the coefficients and consequently, the more accurate are the predictions. The volume of this ellipsoid is inversely proportional to the square root of the determinant of the information matrix [17]. With the D-criterion one tries to maximise this determinant, which is the same as minimising the volume of the ellipsoid.

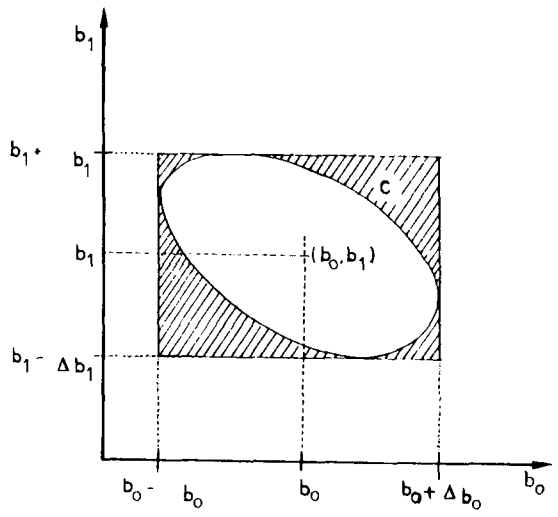


Fig. 2. Confidence intervals of the model coefficients (b) (adapted from Ref. [15]).

Let us consider two matrices, ξ_1 and ξ_2 having the same number of experiments n. $\mathbf{X}_1$ and $\mathbf{X}_2$ are the corresponding model matrices. According to the D-criterion, $\mathbf{X}_1$ will be considered better than $\mathbf{X}_2$ if $\det(\mathbf{X}_1'\mathbf{X}_1) > \det(\mathbf{X}_2'\mathbf{X}_2)$. However, when the matrices do not have the same number of experiments, the comparison of the determinants cannot be done as such. To eliminate the influence of the number of experiments and allow the comparison we use in place of the information matrix $(\mathbf{X}'\mathbf{X})$, a matrix defined as $\mathbf{M} = (\mathbf{X}'\mathbf{X})/n$. This is called the *matrix of moments* and n is the number of rows in the matrix $\mathbf{X}$. A design matrix $\mathbf{X}_{n1}$ will be more efficient than another, $\mathbf{X}_{n2}$, if $\det \mathbf{M}(\mathbf{X}_n)_1 > \det \mathbf{M}(\mathbf{X}_n)_2$.

3.2. A-criterion or trace criterion

This criterion minimises the average variance of the estimated coefficients and a model matrix is considered A-optimal when the trace of the corresponding dispersion matrix is minimum.

$$\operatorname{tr}(\mathbf{X}^{*\prime}\mathbf{X}^{*})^{-1} = \min_{\xi_n \Xi_n} \operatorname{tr}(\mathbf{X}'\mathbf{X})^{-1} \tag{4}$$

The variance of the estimated coefficients is described as

$$\operatorname{Var}(b_i) = c_{ii}\sigma^2 \tag{5}$$

where b_i is the estimation of the coefficient β_i, c_{ii} is the value on the main diagonal of $(\mathbf{X}'\mathbf{X})^{-1}$ and σ^2 is the experimental variance.

The trace of a square matrix, in our case $(\mathbf{X}'\mathbf{X})^{-1}$, is equal to the sum of the p elements on the main diagonal (that is equal to the number of model parameters, since the matrix is square). The σ^2 depends exclusively on the experimental method chosen and is supposed to be equal for all experiments. If we denote $(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{A}$, the trace of $\mathbf{A}$ would be represented as

$$\operatorname{tr}(\mathbf{A}) = \sum_{i}^{p} c_{ii} \tag{6}$$

Using the design matrix above as an example, the trace of $(\mathbf{X}'\mathbf{X})^{-1}$ would be equal to 3.12 and is calculated as follows:

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{pmatrix} 2.99 & -0.57 & -0.00 \\ -0.57 & 0.13 & 0.00 \\ -0.00 & 0.00 & 0.00 \end{pmatrix}$$

$$\operatorname{tr}(\mathbf{X}'\mathbf{X})^{-1} = \sum_{i=1}^{p} c_{ii} = 2.99 + 0.13 + 0.00 = 3.12$$

3.3. Variance function

The variance function or leverage is a measure of the uncertainty in the predicted response [18]. It is described as follows

$$\operatorname{Var}(\tilde{y}_i) = \boldsymbol{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\boldsymbol{x}_i * \sigma^2 = d(x_i) * \sigma^2 \tag{7}$$

where $\operatorname{Var}(\tilde{y}_i)$ represents the variance in the prediction at point i, $d(x_i)$ is the variance function at point i and $\boldsymbol{x}_i$ is a vector describing an experiment.

The ideal situation would be when the variance function is as small as possible, leading to minimum deviations between predicted and true values. When $d(x_i)$ is equal to 1, the variance in the prediction will be the same as the variance of the experimental method (σ^2). This usually is considered acceptable, so that one does not try to find a value smaller than 1.

Let us give an example of how to calculate the variance function ($d(x_i)$) of a point.

Using the model matrix shown before and choosing, for instance, the point number 4 in the matrix of

candidate points (ξ_2), the variance function will be calculated as follows:

$$[1\,5.1\,676] * \begin{pmatrix} 2.99 & -0.57 & -0.00 \\ -0.57 & 0.13 & 0.00 \\ -0.00 & 0.00 & 0.00 \end{pmatrix} * \begin{pmatrix} 1 \\ 5.1 \\ 676 \end{pmatrix} = 0.40$$

$$(x')^{(2)} * (\mathbf{X}'\mathbf{X})^{-1} * x^{(2)} = (dx)^{(2)}$$

where $x^{(2)}$ represents point number 4 described as a column vector, according to a model, $(x')^{(2)}$ is the transpose of $x^{(2)}$ and $(dx)^{(2)}$ is the value of the variance function for point number 4.

In this example, V_{pred} would be calculated as follows:

$$V_{pred} = 0.40 * \sigma^2$$

3.4. G-efficiency

The G-efficiency is described in Eq. (8) and takes into account not only the number of coefficients of the model but also the number of experiments and the variance function.

$$\text{G} - \text{eff} = \left(\frac{p}{d(x) * n} \right) \tag{8}$$

where p is the number of coefficients of the model, $d(x)$ is the largest variance function in the design and n is the number of experiments in the model matrix.

4. Selecting the number of points in the design

The next question we need to answer is *How many points are needed?* A priori there are no rules to define the exact number of experiments one should perform. The minimum number depends on the model proposed. In our example, for instance, the model proposed (Model 1) has seven coefficients which requires us to perform at least seven experiments. The optimal number depends on the constraints imposed by the cost of the experiments, the time and the difficulty to perform them, the precision required, etc. The advantage of using the criteria described above is that one is able to choose the best design for a particular problem *without performing any experiment*. This allows us to find the best design for any number of experiments from the minimum needed to the maximum one is prepared to perform.

Let us start by determining the D-optimal design with seven experiments, the minimum allowed as explained above. To apply the D-criterion we would have to calculate the determinant of the information matrix for each combination of seven points (experiments) among the candidate points. The number of combinations for which D should be determined is

$$C_N^n = \frac{N!}{[n!(N-n)!]} \tag{9}$$

where N represents the number of candidate points and n the number of points in the model matrix.

In our example, N is equal to 257 and n is equal to 7. To try all the possible combinations, we will have to calculate approximately 1.4×10^{13} determinants ($\det(\mathbf{X}'\mathbf{X})$) and choose the largest from them. If we increase the number of experiments in the design, the number of calculations required increases rapidly. It is clear that trying all possible combinations is not the best strategy. Several algorithms were developed in order to find a D-optimal design in a more efficient way. These algorithms do not determine the D-optimal design but an approximate D-optimal design [19]. The first algorithm was proposed by Mitchell [20]. In our example we use the algorithm of Fedorov [15,20]. A discussion about these two algorithms can be found in Ref. [21]. Many other algorithms are based on Fedorov's algorithm and a good review of them can be found in Ref. [19]. This algorithm is also called an exchange algorithm because the basic idea is to replace in the chosen matrix ξ_n, the point that gives least information about the domain studied, by the candidate point, that will add the largest amount of information about the domain. The amount of information yielded by each point i is quantified by the variance function (dx) calculated according to Eq. (7).

In all criteria mentioned above, the model proposed is very important because the selection of the best set of points is based on the model matrix (**X**).

A review about all these methods and models can be found in Refs. [22,23].

5. Adding or deleting experiments

To select an optimal design (i.e. optimal number of experiments and optimal location of those experi-

ments in the experimental domain), we will need to determine the effect of exchanging an experiment for another experiment from such a set [24].

Suppose a starting matrix $\mathbf{X}_n^{(0)}$ containing n experiments. The corresponding information matrix is $(\mathbf{X}'_n\mathbf{X}_n)_{(0)}$ and the variance function (dx) of any candidate point in the experimental domain (x^j) is described as follows

$$d(x^j) = x^{j\prime}(\mathbf{X}'_n\mathbf{X}_n)_{(0)}^{-1}x^j \qquad (10)$$

When one adds a point x^j to the matrix $\mathbf{X}_n^{(0)}$, a new matrix $\mathbf{X}_{n+1}^{(1)}$ is created. For this new matrix the information matrix is $(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)}$.

The relation between these two information matrices can be expressed by

$$(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)} = (\mathbf{X}'_n\mathbf{X}_n)_{(0)} + (x^j * x^{j\prime}) \qquad (11)$$

and the determinants of the corresponding matrices are

$$|\mathbf{X}'_{n+1}\mathbf{X}_{n+1}|_{(1)} = |\mathbf{X}'_n\mathbf{X}_n|_{(0)} * (1 + d(x^j)) \qquad (12)$$

Point x^j is chosen such that its addition leads to the maximum increase of the determinant $|(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)}|$.

In the same way, it is possible to calculate the variance function for any point (x_i) making part of the new matrix $\mathbf{X}_{n+1}^{(1)}$.

$$d(x_i) = x_i'(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)}^{-1}x_i \qquad (13)$$

If one eliminates one of the points (x_i), a new matrix $\mathbf{X}_n^{(2)}$, is formed and its information matrix is expressed by $(\mathbf{X}'_n\mathbf{X}_n)_{(2)}$.

The relation between these two matrices $(\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)}$ and $(\mathbf{X}'_n\mathbf{X}_n)_{(2)}$ is

$$(\mathbf{X}'_n\mathbf{X}_n)_{(2)} = (\mathbf{X}'_{n+1}\mathbf{X}_{n+1})_{(1)} - (x_i * x_i') \qquad (14)$$

and the determinants can be described as

$$|\mathbf{X}'_n\mathbf{X}_n|_{(2)} = |\mathbf{X}'_{n+1}\mathbf{X}_{n+1}|_{(1)} * (1 - d(x_i)) \qquad (15)$$

Point x_i is chosen such that its suppression leads to the minimum decrease of the determinant $|\mathbf{X}'_n\mathbf{X}_n|_{(2)}$.

Eqs. (12) and (15) show that the addition and suppression of points is based on the variance function. The point to be added must have the largest variance function because such a point is the most different from those already selected. The point to be suppressed has the smallest variance function. This is the basis of the algorithm of Mitchell.

6. Fedorov's algorithm

In Fedorov's algorithm the exchange of point x_i from the model matrix with the point x^j from the

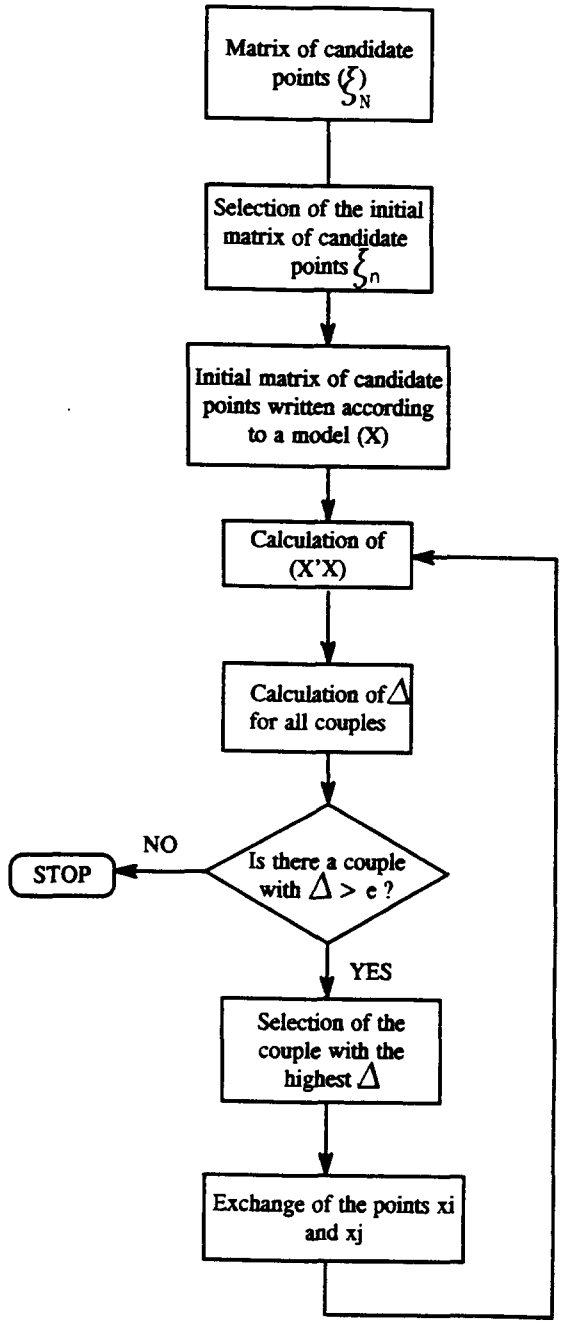


Fig. 3. Flowchart of the algorithm of Fedorov.

matrix of candidate points, is made simultaneously and the new information matrix is related to the previous one by the relation

$$(\mathbf{X}'_n\mathbf{X}_n)_{(1)} = (\mathbf{X}'_n\mathbf{X}_n)_{(0)} - (x_i * x'_i) + (x^j * x^{j\prime}) \tag{16}$$

The corresponding determinants are linked by the relation

$$|\mathbf{X}'_n\mathbf{X}_n|_{(1)} = |\mathbf{X}'_n\mathbf{X}_n|_{(0)} * \left(1 + \Delta(x_i, x^j)\right) \tag{17}$$

where

$$\Delta(x_i, x^j) = d(x^j) - \left[d(x_i)d(x^j) - d^2(x_i, x^j)\right] - d(x_i)$$

$$d(x_i) = x'_i(X'_nX_n)^{-1}_{(0)}x_i,$$

$$d(x^j) = x^{j\prime}(X'_nX_n)^{-1}_{(0)}x^j,$$

and

$$d(x_i, x^j) = x'_i(X'_nX_n)^{-1}_{(0)}x^j = x^{j\prime}(X'_nX_n)^{-1}_{(0)}x_i \times (\text{covariance function}).$$

Starting from a matrix of n points with $n \geqslant p$, selected from the initial matrix of candidate points ξ_N, the first step is the construction of the model matrix $\mathbf{X}$ and the calculation of the determinant of the information matrix $(\mathbf{X}'\mathbf{X})$. As a second step, the algorithm calculates the variance function (dx_i) for each point (x_i) of matrix ξ_n and for each remaining point (x^j) in the matrix of candidate points. The algorithm calculates the Δ value for all possible couples. Once the couple (x_i, x^j) that leads to the maximum $\Delta(x_i, x^j)$ is found, the algorithm exchanges x_i by x^j and recalculates the determinant of the new ξ_n matrix. When there is more than one couple with the same Δ value, the algorithm chooses randomly among them. Theoretically, the D-optimal matrix is found when there is no more couple with positive Δ. However, sometimes the Δ values are practically the same and although it would not make any significant difference in the determinant, the algorithm could continue for a long time. To avoid this a very low threshold value (e) can be established, in such a way that when $\Delta < e$, the algorithm stops and the matrix is considered as optimal. Fig. 3 shows the flowchart of the algorithm.

A numerical example of how this algorithm proceeds is described below. Using the matrix of candidate points (ξ_6), model 2 and the matrix with 4 experiments (ξ_4) as the initial design, the algorithm

[a] creates from ξ_4, the model matrix according to model 2. We represent this matrix as $\mathbf{X}^{(0)}$ to distinguish it from other model matrices that will be developed by the algorithm. This matrix is given below.

$$\mathbf{X}^{(0)} = \begin{pmatrix} 1 & 2.1 & 900 \\ 1 & 3.5 & 225 \\ 1 & 4.9 & 1600 \\ 1 & 5.7 & 100 \end{pmatrix}$$

[b] calculates $\det(\mathbf{X}^{(0)\prime}\mathbf{X}^{(0)})$. For this matrix this is equal to 4.2851×10^7.[c] calculates Δ for all couples (x_i, x^j). In this example there are eight couples, that can be seen in the matrix scheme below.

		$\mathbf{X}^{(0)}$			
		1	2	3	5
ξ_2	4	$\Delta(x_1,x^4)$	$\Delta(x_2,x^4)$	$\Delta(x_3,x^4)$	$\Delta(x_5,x^4)$
	6	$\Delta(x_1,x^6)$	$\Delta(x_2,x^6)$	$\Delta(x_3,x^6)$	$\Delta(x_5,x^6)$

In the scheme each column represents a point in the design matrix $(\mathbf{X}^{(0)})$, that is points number 1, 2, 3 and 5, and each row, a point in the matrix of candidate point (ξ_2), that is points 4 and 6. The elements in the matrix scheme represent a couple formed with one point from each matrix ($\mathbf{X}^{(0)}$ and ξ_2). For example, the couple formed by point number 5 in $\mathbf{X}^{(0)}$ and point number 4 in ξ_4 is described by $\Delta(x_5, x^4)$ and can be interpreted as an exchange of point 5 by point 4, in the design matrix $(\mathbf{X}^{(0)})$.

In this procedure we exchanged a point from $\mathbf{X}^{(0)}$ with a point from ξ_2. This is called an exhaustive search (E). Another way of finding the points to be exchanged is called non-exhaustive search (NE) [21]. In this approach the point to be exchanged with the point from $\mathbf{X}^{(0)}$ does not come from ξ_2 but from ξ_6. The consequence is that instead of having four different points in the new design matrix, it is possible that the exchange will result in a matrix where an ex-

periment is performed twice. In this paper we will base our example on the exhaustive search but the general reasoning applied in both approaches is in general the same.

The values of Δ calculated for the couples can be seen below.

		$X^{(0)}$			
		1	2	3	5
ξ_2	4	-0.6659	0.2338	-0.8883	-0.5085
	6	-0.1643	0.2716	-0.5589	0.2569

[d] selects among the couples, the one with the highest value of Δ. In our example it is the couple (x_2, x^6).[e] exchanges point 2 from ξ_4 by the candidate point number 6. Now we have a new matrix ξ_4 formed by the points 1, 3, 5 and 6. The corresponding model matrix ($\mathbf{X}^{(1)}$) is then built.

$$\mathbf{X}^{(1)} = \begin{pmatrix} 1 & 2.1 & 900 \\ 1 & 4.9 & 1600 \\ 1 & 5.7 & 100 \\ 1 & 7.0 & 576 \end{pmatrix}$$

[f] goes back to step [b] and calculates the determinant for the new design matrix ($\det(\mathbf{X}^{(1)\prime}\mathbf{X}^{(1)})$). Now it is equal to 5.4484×10^7. One can see that the determinant has indeed increased and that the exchange indeed improves the design.[g] the algorithm continues with step [c]. Now the Δ values will be calculated for the couples ($\Delta(x_1, x^2)$, $\Delta(x_1, x^4)$, $\Delta(x_3, x^2)$, $\Delta(x_3, x^4)$, $\Delta(x_5, x^2)$, $\Delta(x_5, x^4)$, $\Delta(x_6, x^2)$, $\Delta(x_6, x^4)$).The results are given below.

		$X^{(0)}$			
		1	3	5	6
ξ_2	2	-0.3428	-0.6529	-0.0113	-0.2134
	4	-0.8218	-0.7942	-0.4626	-0.3974

This time there is no couple with $\Delta > 0$ and the algorithm stops. The design matrix ($\mathbf{X}^{(1)}$) is then the D-optimal design.

7. Initialization

In the previous example the matrix (ξ_4) was randomly chosen. If we start from another set of experiments, for instance experiments number 2, 3, 4 and 6, the optimal design matrix will not necessarily be the same. This happens because we are not calculating the optimal design but an approximation of it, and it is possible to find different solutions for the same problem. They will however have similar determinants (D values). The choice of the initial design matrix (the initial set of experiments) will influence mainly the number of cycles the algorithm will have to perform before it finds an optimal solution. There are several ways of finding an initial solution. The simplest one is a random selection. If the initial design matrix is already close to the optimal design matrix, the number of calculations will be significantly reduced. The other ways of finding an initial design are based on algorithms such as that of Galil and Kiefer [25]. We prefer the Kennard and Stone algorithm [26]. The basic idea of this algorithm is to find the most homogeneous distribution of points in the variable space. This is done using the Euclidean distance as the criterion to select the set of experiments. A good example of the use of the algorithm of Kennard and Stone to find a design is described in Ref. [27].

8. The case study

Using the algorithm of Fedorov, we determined the D-optimal design with seven to fourteen experiments (points) from the matrix of candidate points (ξ_N). The results are presented in Fig. 4(a–d). The first plot (Fig. 4a) describes the behaviour of the logarithm of the determinant of the matrix of moments (**M**) as a function of the number of experiments. One can see that when the number of experiments increases from seven to ten, the log det(**M**) increases gradually. After ten however, it decreases slowly to start increasing again when the number of experiments is equal to fourteen. As fourteen experiments is considered to be too much, we can say that a de-

sign with nine or ten experiments would be the best one. In plot 4b we can follow the behaviour of the trace of the dispersion matrix (A-criterion). The smaller the trace, the more reliable are the calculated coefficients of the model. It should be stressed that in this example we are not trying to find the A-optimal design but verifying how the trace evolves for the D-optimal designs suggested. One can see that increasing the number of experiments from seven to ten makes the trace decrease rapidly. The third plot (Fig. 4c) represents the evolution of the largest variance function when the number of experiments increases. The solid line in the plot represents a variance function equal to one. In our example, a design with seven experiments has a very large variance function. Including more experiments it decreases and a design

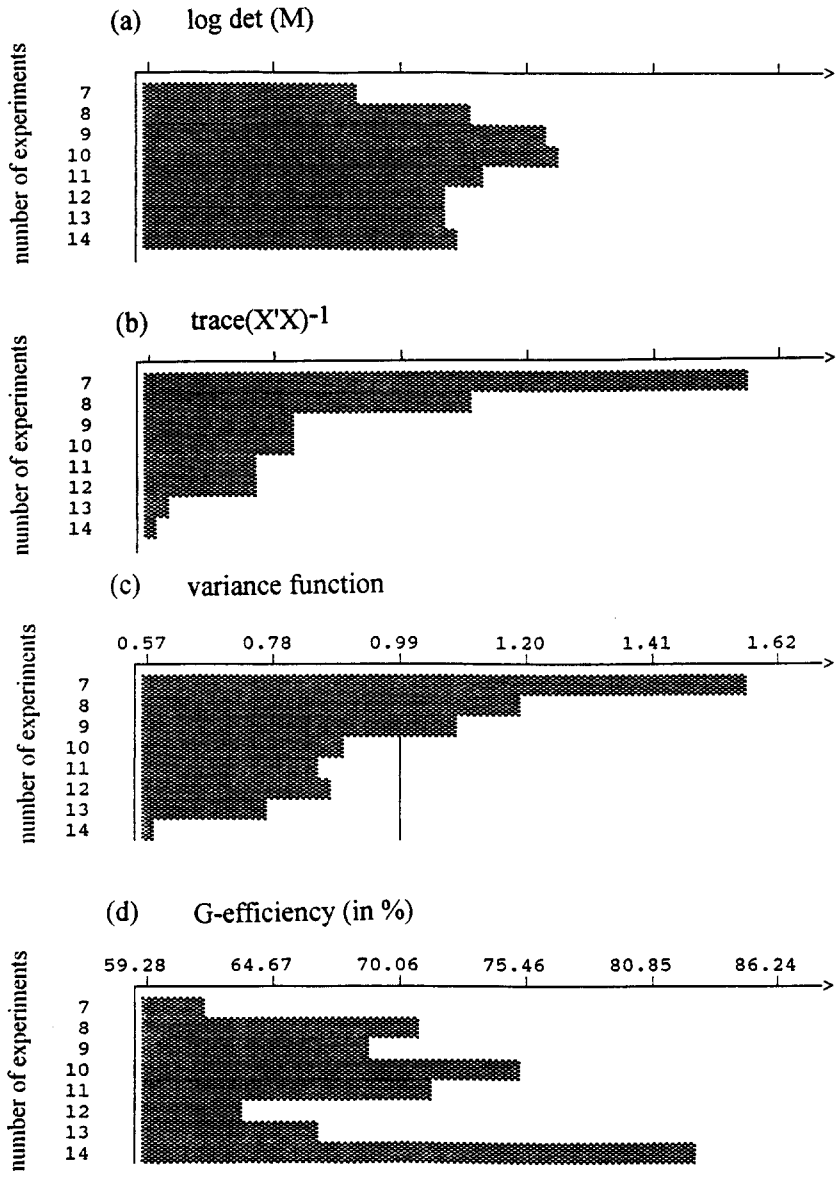


Fig. 4. Plots of (a) log det(**M**), (b) trace $(\mathbf{X}'\mathbf{X})^{-1}$, (c) variance function and (d) G-efficiency, in %, as a function of the number of experiments.

Table 2
Values of D-, A-, $d(x)_{max}$ and G-eff-criteria for the design with ten experiments

log det(**M**)	2.069
trace $(\mathbf{X}'\mathbf{X})^{-1}$	0.917
$d(x)_{max}$	1.3×10^{6}
G-eff (%)	76.34

with nine or ten experiments would be the best. The last plot (Fig. 4d) shows the efficiency of each of these designs. In general, increasing the number of experiments will increase the efficiency of the design and the design with fourteen experiments would be the best. However, the difference in efficiency between the design with ten and fourteen experiments is not very large and does not justify performing four experiments more.

Looking only to the graphics in Fig. 4, these designs (with nine and ten experiments) are equivalent according to the D-, A- and variance function criteria. For the efficiency criterion the design with ten experiments seems to be slightly better.

The points suggested for both designs are shown in Fig. 1. One can see that for one design the points are spread on the boundaries of the region. However, including one experiment more, it would be placed inside the experimental region. Based on this fact, this design would be advised.

Table 2 shows the values of D-, A-, G-eff- and the maximum variance function for this design.

9. Applications of D-optimal designs

The method described in this paper was applied for the separation in HPLC. Compared to the non-linear model proposed by Schoenmakers [27], the combination of Model 1 and the approach proposed in this paper shows that the results of this last are as good as the ones obtained with the first, which avoids the problem of choosing the starting values inherent to non-linear regressions.

The usual symmetrical designs (factorial, central composite, etc.) are D-optimal. It is then of course not necessary to compute D. The calculations connected with the D-optimality concept are carried out, in conjunction with those for the other optimality criteria, when, for some reason, one cannot apply the usual designs. In our case study there are two reasons: the symmetrical designs do not fit the irregular experimental domain and are not suited for a third-order model. Particularly irregular experimental boundary problems occur quite often in chemistry. A typical simple situation is the following. One wants to carry out an organic synthesis and is interested in the effect of the percentage of two reagents. One can add, say, a maximum of 5% of A and B. Therefore the highest level in the design would be 5% for A and 5% for B. It often happens, however, that one cannot add so much if the two reagents are added at the same time. The resulting experimental domain will be irregular as shown in Fig. 5.

Another typical situation occurs when the experimenter cannot control the levels. Such a situation can again be illustrated with an example from organic synthesis [28]. In this example one wants to study the rate of a reaction of tertiary amines with different substituents in function of two physicochemical parameters, namely the inductive effect and the resonance effect. The experimental domain is not symmetrical; it is limited by the number of substituents. The difficulty comes from the fact that it is not possible to choose combinations of these two parameters, in other words, it is not possible to define a range of variation for these parameters; the domain is composed by discrete points in the space. What one can choose is the nature of the substituents and consequently, the levels (values) of each parameter is automatically selected.

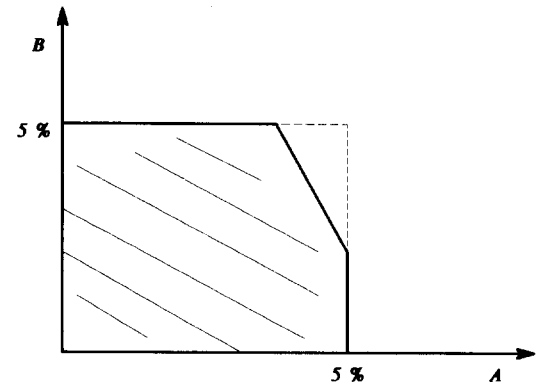


Fig. 5. Experimental region limited by the constraints imposed by some combinations of variables *A* and *B*.

Three models were proposed to link the reaction rate to the parameters and what one wants is to perform the least experiments possible to achieve his aim. Using the approach described in this paper it was possible to determine a reasonable number of experiments to be performed and to select which model was the best for the purposes established by the experimenters. Moreover, inherent to the selection of the experiments the number of substituents that should be taken into account in this study was also determined.

10. Program

All calculations were performed by NEMROD software [29] and MATLAB for Windows version 4.0 [30], both for IBM compatibles.

11. Nomenclature

β	true coefficient of a proposed model
b	coefficient of the model
dx	variance function of a point x
ξ_N	matrix of candidate points
ξ_n	matrix chosen from ξ_N
$\mathbf{M}$	matrix of moments
n	subset of points selected from N
N	number of candidate points
η	true value of the analytical response
p	number of the model coefficients (parameters)
V	variable of the model
x	vector representing one experiment
x'	transpose of x
$\mathbf{X}$	design matrix
$\mathbf{X}'$	transpose of $\mathbf{X}$
$\mathbf{X}^*$	optimal design matrix
$(\mathbf{X}'\mathbf{X})$	information matrix
$(\mathbf{X}'\mathbf{X})^{-1}$	dispersion matrix
Y	dependent variable of the model

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